Jacob Patrick Noe

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Software engineer with 4+ years of experience building and operating production web and mobile systems. End-to-end ownership across Python, TypeScript, PostgreSQL, Redis, AWS, and Docker; promoted to lead developer in 2025. Additional experience developing applications for university faculty and independent projects involving longitudinal behavioral data, associative memory retrieval, and sensor-driven interfaces.

Technical Skills

Languages Python, JavaScript, TypeScript, SQL, C, C++, Kotlin, Swift

Applications and data Flask, FastAPI, React, React Native / Expo, PostgreSQL, MySQL, Redis, D3.js, Socket.IO

Engineering Git / GitHub, Docker, AWS, CI/CD pipelines, offline-first architecture, real-time synchronization, automation and scripting

Professional Experience

Hero Services | Subsidiary of Ellison Environmental

Lead Full Stack Developer, 2025-Present | Full Stack Developer, 2022-2025

- Assumed end-to-end ownership of a production platform supporting water and wastewater facility operations across multiple client organizations.
- Designed offline-first React Native / Expo workflows with durable local storage, network-aware request queuing, and synchronization for unreliable field connectivity.
- Implemented real-time synchronization between React clients and Flask using Socket.IO, with Redis-backed room membership and resource locks for coordinating concurrent users.
- Maintained dynamic forms generated from checksheet specifications, custom D3 visualizations, reporting systems, and data-processing features.
- Maintained scheduled and event-driven email workflows for checksheet due states, completion events, and configurable out-of-range reading alerts.
- Improved application security through a password-hashing migration and production authentication updates.
- Maintained branch-triggered AWS CodeBuild pipelines with component-specific frontend and API/Lambda builds. Lambda updates deployed automatically; Docker-based frontend and API services required manual ECS restarts to complete releases.

California Polytechnic State University

Full Stack Developer | Dr. Brennan Davis | 2020-2021

- Served as sole developer, leading the design and implementation of public-facing websites for university faculty.
- Developed interfaces, database architecture, and API integrations using JavaScript, React, Python, PHP, and MySQL; managed domain configuration.

iOS App Developer | Dr. Brennan Davis and Dr. Suzanne Phelan | 2020

- Developed an augmented-reality experience and a diet-tracking application using Swift and Unity / C#.
- Integrated barcode scanning, nutrition APIs, and 3D coordinate systems into mobile user experiences.

Selected Projects

ORBIT | Longitudinal behavioral logging and exploratory clustering

- Built a functioning mobile application for logging mood, activities, diet, exercise, and medication over time.
- Implemented clustering over state vectors representing current and recent observations to explore patterns across logged mood and behavioral data; recommendation functionality remains in development.

AI Memory Harness | Associative retrieval and memory graphs

- Built a memory-retrieval harness combining semantic matching with recursive traversal of potentially relevant memories; strengthened graph links between memories retrieved together.
- Implemented a separate emoji-based graph to represent emotional associations, balancing semantic and emotional retrieval signals with recency and retrieval-derived memory groups.

WHORL | Sensor-driven musical interface

- Integrated an Arduino GIGA R1, Hall sensor, and magnet-equipped fidget spinner into a working controller for synthesizer modulation in Logic / Alchemy.
- Used C++ firmware for pulse tracking, RPM estimation, smoothing, and stopping detection, with a Python serial-to-MIDI bridge; documented the protocol and added automated parser tests through AI-assisted development.

Additional Experience

Shopify Developer | From War to Peace | 2025

- Worked within an existing Shopify website, integrating apps and implementing custom solutions in Liquid.

Full Stack Developer | Early-Stage Startup | 2021

- Co-founded a city-funded startup focused on improving access to employment pathways for unhoused individuals.
- Designed and built a D3 node-graph interface for exploring employment opportunities.

Education

California Polytechnic State University

Bachelor of Science in Computer Science | December 2021

Cum Laude | GPA 3.56